

JSCR-08-4326 R-1

**MULTIARTICULAR ISOKINETIC HIGH LOAD ECCENTRIC TRAINING INDUCES
LARGE INCREASES IN ECCENTRIC AND CONCENTRIC STRENGTH AND
JUMPING PERFORMANCE**

Konstantinos Theodosiou¹, Gregory C. Bogdanis², Evangelia Gkantiraga¹, Ioannis Gissis¹,
Michalis Sambanis¹, Athanasios Souglis², and Aristomenis Sotiropoulos², Christos
Papadopoulos¹,

¹Laboratory of Sport Biomechanics, Faculty of Physical Education and Sports Science at Serres,
Aristotle University of Thessaloniki, and ²Faculty of Physical Education and Sports Science,
University of Athens, Greece.

Brief running head: Eccentric training increases eccentric force and jumping performance

Christos Papadopoulos, PhD,

Sport Biomechanics Laboratory, Department of Physical Education and Sports Science, Aristotle
University of Thessaloniki, Agios Ioannis, 62110 Serres, GR,

Telephone number: +30 2310 991077,

Fax number: +30 2310 991078,

e-mail: chrispap@phed-sr.auth.gr

ABSTRACT

This study investigated the effects of short-term eccentric exercise training using a custom made isokinetic leg press device, on concentric and eccentric strength and explosiveness as well as jumping performance. Nineteen healthy males were divided into an eccentric (ECC, n=10) and a control group (CG, n=9). The ECC group trained twice per week for 8 weeks using an isokinetic hydraulic leg press machine against progressively increasing resistance ranging from 70 to 90% of maximal eccentric force. Jumping performance and maximal force generating capacity were measured before and after eccentric training. In the ECC group, drop jump (DJ) height and maximal power were increased by $13.6 \pm 3.2\%$ ($p < 0.01$) and $25.8 \pm 1.2\%$ ($p < 0.01$), while ground contact time was decreased by $17.6 \pm 2.6\%$ ($p < 0.01$). Changes in, ankle, knee and hip joint angles were also reduced by $33.9 \pm 1.1\%$, $31.1 \pm 1.0\%$ and $32.4 \pm 1.6\%$ (all $p < 0.01$), respectively, indicating an increase in muscle stiffness during the DJ. Maximal eccentric and concentric leg press force was increased by $64.9 \pm 5.5\%$ ($p < 0.01$) and $32.2 \pm 8.8\%$ ($p < 0.01$), respectively, and explosiveness, measured as force attained in the first 300 ms, was increased by $49.1 \pm 4.8\%$ ($p < 0.01$) and $77.1 \pm 7.7\%$ ($p < 0.01$), respectively. The CG did not show any statistically significant changes in all parameters measured. The main findings of the present study were that maximal concentric and eccentric force, explosiveness and DJ performance, were markedly increased after only 16 training sessions, possibly due to the high eccentric load attained during the bilateral eccentric leg press exercise performed on this custom made device.

Key Words: leg press, drop jump, muscle stiffness, force.

INTRODUCTION

Eccentric training is a popular exercise modality that is used by healthy and injured athletes to improve muscle strength and explosiveness (20). Due to the high muscle force combined with low energy cost of eccentric vs. concentric contractions (9), eccentric training has been administered to older adults, resulting in increases in strength and functionality (24). Eccentric or plyometric training is frequently used by speed and power athletes aiming to improve muscle strength, explosiveness and jumping performance (11,34).

Eccentric training may be considered superior to concentric in several aspects. A meta-analysis of studies comparing eccentric and concentric training by Roig et al. (34) showed that eccentric training results in greater strength gains, due to the fact that exercise intensity is higher. High mechanical tension is an important factor influencing muscle hypertrophic responses which are, in turn, related with increases in muscle strength (37). During eccentric contractions, passive muscular tension is also increased due to the lengthening of the passive muscle elements containing collagen and titin (38). This augments the active tension developed by the contractile elements, enhancing the hypertrophic response (37). The greater hypertrophic response has also been documented by results from human studies showing greater skeletal muscle protein synthesis following bouts of maximal eccentric than concentric exercise (29).

Neural adaptations are also different in eccentric compared with concentric exercise training, possibly due to differences in muscle activation and recruitment patterns between the two types of exercise (13). Hortobagyi et al (18) reported that EMG activity was increased by 86% during eccentric testing following eccentric training, while EMG activity during concentric

testing was increased by only 12%. This seven-fold greater improvement of EMG activity after eccentric compared with concentric training was coupled with a 3.5 times greater increase in eccentric strength (46 vs. 13%). Moreover, eccentric training results in a selective recruitment (28) and hypertrophy of fast twitch-type II motor units (18), which may contribute to the greater effectiveness of this type of training.

In sports practice, eccentric strength training may be performed in combination with plyometric or drop jump training. This type of training is a popular and effective method for improving jumping performance and is widely used among power athletes (5,35). Drop jumps involve a double leg eccentric muscle contraction, as the body drops from a height, followed by a forceful concentric contraction, performed as fast as possible (“stretch-shortening cycle”). Recent studies show that a high level of leg strength is important for drop jump performance, since it enables the athlete to use greater drop jump heights, which increase the training stimulus (5,31). In that respect, eccentric training may be an effective means to induce large increases in muscle strength and jumping performance, when it precedes or is combined with drop jump training. However, most studies using eccentric exercise training have used open-chain single limb movements, e.g. leg extension (17,18,39) that do not train muscles in a sports task-specific manner. In contrast, when closed-chain exercises are used, e.g. squat (11) or Olympic lifts (3), the movement is closer to sports movements (e.g. jumping), but the spine is heavily loaded and the possibility of injury is increased. Thus, it would be desirable to be able to use high eccentric loads using closed-chain movements, while minimizing the load on the spine. This may be useful for athletes who are not as strong in the upper body (e.g. females) or do not wish to load their upper body with unnecessary load (e.g. injured upper body). Also, this may be desirable for

elderly individuals who wish to increase their lower body strength and maintain their fast twitch fibres (24).

Therefore, the purpose of the present study was to examine the effects of high load eccentric training on concentric and eccentric muscle strength and explosiveness, as well as on jumping ability. A custom made isokinetic leg press hydraulic machine was used with the subject in the seated position, to minimize loading on the upper body. Furthermore, to reduce the possibility of injury or extensive muscle damage and to make the movement more task-specific, the range of motion of the leg press movement was confined to approx. 30° at the knee joint (from 125° to 155°, where 180° refers to straight knee). It was hypothesized that the combination of eccentric contractions with high resistance throughout a task-specific range of motion, would elicit large increases in strength and jumping performance.

METHODS

Experimental Approach to the Problem

To evaluate the effects of short-term, high load eccentric training, 19 recreationally active men were randomly assigned into an eccentric training group (ECC group) and a control group (CG group). Following familiarization with the testing procedures and equipment, all subjects performed the baseline tests. Then, subjects in the ECC group trained twice per week for 8 weeks using a custom made isokinetic hydraulic leg press machine, against progressive resistance ranging from 70 to 90% of maximal eccentric force. Subjects in the control group continued their recreational activities and were asked not to get involved in any type of resistance exercise for the duration of the study. The following dependent variables were measured before (pre-) and after (post-) training: (a) On the isokinetic leg press machine: maximal concentric ($F_{\max\text{CON}}$) and

eccentric force ($F_{\max\text{ECC}}$) as well as the force attained in the first 300 ms of the concentric and eccentric max efforts ($F_{300\text{CON}}$ and $F_{300\text{ECC}}$, respectively), (b) During a drop jump (DJ) from a height of 0.3 m: maximal jump height ($\text{DJ}_{\text{height}}$), support time (DJ_{ts}) maximal power (DJ_{Pmax}), changes of the ankle, knee and hip joint angles. The pre and post-training tests were performed at the same time of the day, while diet was recorded for 2 days before each test during the pre-training period and replicated for the 2 days preceding the post-training tests.

Subjects

Nineteen healthy male students (age: 21.3 ± 0.9 years; body mass: 78.3 ± 8.0 kg; height: 180.0 ± 6.7 cm) with no history of musculoskeletal injuries, took part in this study. After a thorough description of the protocol, risks and benefits of the study and the right to terminate participation at will, subjects gave their written consent. None of the participants had engaged in systematic strength training for at least 6 months before the start of the study, but they were active in recreational sports. Approval for the project was obtained from the committee on human research of the Aristotle University of Thessaloniki. All procedures used in this study were in conformity with the Declaration of Helsinki.

Equipment

The isokinetic leg press hydraulic machine was developed in collaboration with Hydrodynamic North Greece (Commercial & Industrial S.A., Thessaloniki, Greece). It uses an electric motor (CMS Motor, 15HP, 1430 rpm, 3-phase electric motor, 380V, Borgo Val di Taro, Italy), an oil tank (200 liter) and a plunger on which a support metal base is connected. A tri-axial force plate (OR-6-6-4000, AMTI Inc., Watertown, MA, USA) was fitted on this support

base (Fig. 1). Displacement, velocity and resistance during leg press movements are controlled by two regulators: the flow controller, which regulates linear velocity of the support base (speed range: 0.15-0.75 m/sec) and the pressure regulator, which adjusts resistance using hydraulic pressure (the maximum resistance is about 12000N; at a slow linear velocity of 0.20 m·s⁻¹). Data were collected using the Net Force acquisition software (Version 2.1, 2005, AMTI Inc., Watertown, MA, USA) interfaced with a computer.

 Insert Figure 1 about here

Eccentric Training program

The training program included 16 sessions of eccentric exercise (the concentric phase was performed passively) over an 8-week period. The training program was supervised by one of the investigators. Each training session began with a standardized warm-up, consisting of 10-min cycling at a self-selected pace (range: 60-80 rpm) on a stationary cycle ergometer at a constant power of 100W, 5 min of static and dynamic stretching of the quadriceps, hamstrings, and triceps surae muscles and 3 min of rest. Depending on the progression of the program, a 3-7 min rest was allowed between each set (Table 1). The training load ranged between 3 and 6 sets of 5-10 high load eccentric contractions at (70-90% of the maximal eccentric force).

Training load was increased progressively (14,21), starting from an intensity of 70-80% of maximal eccentric strength in weeks 1-3 (first 6 sessions) and reaching 90% of maximal eccentric strength in weeks 4-8 (in the remaining 10 sessions). This was done to reduce muscle damage and delayed onset muscle soreness (33). Subjects were assisted to maintain the desired force levels by visual feedback, using a video projector that showed the force signal and the

target force during each repetition. Training load was adjusted every two weeks by measuring maximal eccentric force at the beginning of the first session of each 2-week block (weeks 3, 5 and 7).

 Insert Table 1 about here

Testing

Isokinetic maximal concentric and eccentric leg press force

Following at least 3 familiarization sessions, maximal eccentric leg press force was measured at baseline (pre- training) and 3-4 days after training (post-training), as well as three times during the 8-week training (start of weeks 3, 5 and 7) to adjust the training load. Maximal concentric leg press force was measured only at baseline (pre-training) and 3-4 days after training (post-training). To avoid a possible effect of diurnal variations on maximal strength, the measurements were completed at the same time of the day

Following the standardized warm-up (10-min cycling at 100W, 5 min of static and dynamic stretching, and 5 min of rest), subjects were placed on the seat with the backrest reclined by 20° (Fig. 1). The pelvis was secured by a safety belt and two shoulder straps with pads to minimize upper body movement. The footrest was rotated 10° from the vertical towards plantar flexion and the feet were placed 0.10 m above the seat (Fig. 1). Each contraction (concentric or eccentric) was isokinetic and lasted ~700ms. The knee and hip joint were extended (concentric) or flexed (eccentric) with maximal effort. The range of motion of the knee joint was $29 \pm 2^\circ$ (from $126 \pm 1^\circ$ to $155 \pm 2^\circ$, where 180° refers to the straight knee) and the linear velocity

of the foot plate was $0.20 \text{ m}\cdot\text{s}^{-1}$. This resulted in an average isokinetic angular velocity of the knee joint of $\sim 44^\circ \text{ s}^{-1}$ (or 0.77 rad s^{-1}). The angular velocity of the knee joint was determined from video analysis, as detailed below. In each testing session, subjects performed three concentric and three eccentric maximal trials with a 3 min recovery in between. Subjects were verbally encouraged to apply their maximal concentric or eccentric force, as quickly as possible. Visual feedback of the force signal and the maximum values attained was provided both during and after each repetition. The best eccentric and concentric trials were kept for subsequent analysis. Maximal concentric (F_{maxCON}) and eccentric force (F_{maxECC}) as well as the force attained in the first 300 ms of the concentric and eccentric max efforts ($F_{300\text{CON}}$ and $F_{300\text{ECC}}$, respectively), were recorded. The intraclass correlation coefficients (ICC) for the maximal concentric and eccentric force were 0.97 and 0.98 ($p < 0.01$), respectively, and the ICCs for the $F_{300\text{CON}}$ and $F_{300\text{ECC}}$ were 0.94 and 0.93 ($p < 0.01$).

Drop jump performance test

After 20 min of rest following the leg press isokinetic force measurements, subjects performed the standardized warm-up and performed three maximal drop jumps (DJ) from a height of 0.3 m on a force platform (9281 CA, 1000Hz, Kistler Instruments Corp., Winterthur, Switzerland). A 0.3 m wooden box was placed 0.1 m behind the rear edge of the force plate and subjects began the movement from a standardized starting position with the hands held on the hips and the feet aligned along the front edge of the wooden platform. Participants were instructed to fall down, land simultaneously on both feet and complete a rapid, maximal, double-leg jump effort to jump as high as possible. Two or three practice repetitions were performed followed by three main attempts, separated by 2-3 min. The **highest** jump was kept for further

analyses. Maximal jump height (DJ_{height}), the duration of the feet contact with the ground or support time (DJ_{ts}) and the maximal power (DJ_{pmax}) were recorded. The ICCs for jump height, maximal power and support time were 0.98, 0.98 and 0.94 ($p < 0.01$). Also, changes of the ankle, knee and hip joint angles were measured using two-dimensional video analysis. The change in each of these angles was defined from the start (ground contact) until the end of the eccentric contraction during landing. The ICCs for the lower limb angle measurements ranged between 0.95 and 0.98 ($p < 0.01$).

Kinematic analysis

A video camera (JVC-GR-DVL 9800, NTCS, 120Hz) was used for measuring changes in the ankle, knee and hip angles during the isokinetic leg press and the DJ performance tests. Reflective markers with 20 mm diameter were placed on the right side of the body at the following anatomical landmarks: acromion (shoulder), greater trochanter (hip), lateral epicondyle of the femur (knee), lateral point of fibular malleolus (ankle), distal posterior surface of calcaneus (heel), and fifth metatarsal head (foot). Furthermore, one marker visible in the field of view of video camera was attached to the force plate to assess its kinematics (position, time and linear velocity). All data were synchronized and collected at sampling rates of 1000Hz (forces) and 120Hz (kinematic data). Digitization of these points was performed automatically using the Ariel Performance Analysis System (Ariel Dynamics, Inc., San Diego, CA, USA). A low-pass digital filter with a cut-off frequency of 6 Hz, which was chosen after residual analysis for a wide range of cut-off frequencies (43), was used for smoothing the raw position–time data. Marker and force data were further analyzed using custom- written code (Matlab, version 7.5,

The Mathworks Inc, Natick, MA, USA) and commercial software for biomechanical analysis (Ariel Dynamics, Inc., San Diego, CA, USA).

Statistical Analyses

Statistical analyses were performed with Statistical Package for the Social Sciences (version 17.0; SPSS Inc., Chicago, IL). The data were assessed for normality, using the Kolmogorov–Smirnov test. A two-way ANOVA [group (ECC and CG) x time (pre- post-training)] with repeated measures on one factor (group) was used to examine the effects of eccentric training on maximal eccentric and concentric leg press force and DJ performance. Tukey’s post-hoc tests were performed when a significant main effect or interaction was obtained ($p < 0.05$) to locate differences between means. Effect size for main effects and interaction was estimated by calculating partial eta squared (η^2) values. Effect sizes were classified as small (0.06), medium (0.14) and large (> 0.14). Test–retest reliability for all the dependent variables measured in this investigation was determined in separate experiments by calculating the intraclass correlation coefficient (ICC) using a 2-way mixed model. Statistical significance was accepted at $p < 0.05$.

RESULTS

Drop jump performance

The two-way ANOVA showed a significant group x time interaction for DJ_{height} ($p = 0.001$, $\eta^2 = 0.63$). The post-hoc test showed that only the ECC group increased their DJ_{height} by $13.6 \pm 3.2\%$ ($p < 0.001$), while performance of the CG remained unchanged (Table 2). Similarly, there was a group x time interaction for DJ_{ts} ($p = 0.001$, $\eta^2 = 0.68$) and DJ_{Pmax} ($p = 0.001$, $\eta^2 = 0.93$).

Post-hoc tests showed that the ECC group decreased DJ_{ts} by $17.6 \pm 2.6\%$ ($p < 0.001$) and increased DJ_{pmax} by $25.8 \pm 1.2\%$ ($p < 0.001$, Table 2). No changes were observed in the CG for all DJ parameters examined.

Changes in the ankle, knee and hip joint angles are shown in Fig. 2. The two-way ANOVA revealed significant group x time interaction for the change in ankle ($p = 0.001$, $\eta^2 = 0.88$), knee ($p = 0.001$, $\eta^2 = 0.87$) and hip angle ($p = 0.001$, $\eta^2 = 0.78$).). As shown in Fig. 2, ankle, knee and hip joint angles were smaller after training by $18-22^\circ$ or $33.9 \pm 1.1\%$, $31.1 \pm 1.0\%$ and $32.4 \pm 1.6\%$ ($p < 0.01$), **respectively**. Changes in all leg angles were similar in the post- and the pre-training test for the CG.

Insert Fig. 2 about here

Maximal isokinetic eccentric and concentric force

A significant group x time interaction was found for F_{maxECC} ($p = 0.001$, $\eta^2 = 0.90$) and F_{300ECC} ($p = 0.001$, $\eta^2 = 0.84$). Post-hoc tests showed that F_{maxECC} increased by $64.9 \pm 5.5\%$ ($p < 0.01$) and F_{300ECC} increased by $49.1 \pm 4.8\%$ ($p < 0.01$) only in the ECC group (Table 3).

A significant group x time interaction was found for F_{maxCON} ($p = 0.001$, $\eta^2 = 0.33$) and F_{300CON} ($p = 0.001$, $\eta^2 = 0.79$). Post-hoc tests showed that F_{maxCON} increased by $32.2 \pm 8.8\%$ ($p < 0.01$) and F_{300CON} increased by $77.1 \pm 7.7\%$ ($p < 0.01$) only in the ECC group (Table 4). No changes were observed in the CG for all force parameters examined.

At baseline, the ratio of $F_{\max\text{ECC}}$ to $F_{\max\text{CON}}$ was $94.3 \pm 1.6\%$ and $92.5 \pm 7.6\%$ for the CG and ECC groups, respectively. This ratio remained unchanged for the CG ($91.6 \pm 3.5\%$) but was significantly increased for the ECC group to $113.2 \pm 3.6\%$ ($p=0.02$, $\eta^2=0.28$).

Insert Tables 3 and 4 about here

ACCEPTED

DISCUSSION

The main finding of the present study was that high intensity eccentric training using the custom-made isokinetic leg press machine resulted in large increases in maximal eccentric and concentric strength (by 65 and 32%, respectively), explosiveness (by 49 and 77%, respectively) drop jump performance, as quantified by drop jump height and power and ground contact time (by 14, 26 and 18%, respectively). An important aspect of this study was that these neuromuscular adaptations were attained with only 16 training sessions, each composed of 3-6 sets of 6-10 repetitions, totaling just 30-40 repetitions or 20-30 seconds of pure exercise time in each training session. The high eccentric load throughout the range of motion, that was attained using our custom-made isokinetic leg press machine, in combination with the visual feedback that assisted in maintenance of high effort, may partially explain the high efficiency of training observed in the present study.

One advantage of training using eccentric leg press exercise, instead of single-leg, open-chain exercises such as leg extension, is that muscles are activated and loaded in a more sports task-specific manner. Strength and power gains can be readily transferred to athletic movements such as jumping, sprinting and power lifting. Furthermore, the nature of the exercise (i.e. seated leg press), minimizes loading of the spine, thus allowing the use of very high resistance applied safely on the leg muscles. This constitutes an important advantage of this training device and it may be speculated that it could be used to attain high eccentric loading of the leg muscles in a wide range of populations that could not otherwise be able to do so (e.g. elderly, females, injured). The results of the present study can be compared with those of a recent study (27), in which subjects were standing on a platform that was moving up and down at 2.5 Hz (movement

amplitude: 11.4 cm), with a horizontal stationary bar fixed on their shoulders. Subjects were instructed to apply a downward force that was around 70% of the maximal concentric 1RM strength, thus performing both concentric and eccentric muscle actions with their leg muscles (27). Following 10 weeks of training, drop jump performance, explosive force and peak power were increased by 6%, 22.5% and 5%, respectively, while in the present study the magnitude of improvement in the same parameters was more than 2-fold higher. This may be due to the higher load used in the present study (90% of maximal eccentric force vs. 70% of the maximal concentric 1-RM), highlighting the importance of high loading during eccentric training aiming to maximize strength gains. Even when high loads (e.g. 90% of concentric 1RM) are used during traditional half-squat training (7), the magnitude of strength improvement is significantly lower (~18%) compared with the present results. The superiority of eccentric exercise for strength gains and muscle fiber hypertrophy has been highlighted in recent studies and meta-analyses (29,34,37).

One interesting finding of the present study was the large increase in explosiveness, which was assessed by the force attained in the first 300 ms of contraction. This is equivalent to the rate of force development (RFD), which has been shown to be related with muscle fiber type and myosin heavy chain composition (15), muscle fibre cross-sectional area and neural drive to the muscle (1). An increased percentage and cross sectional area of fast twitch fibres is a desirable adaptation to enhance strength and explosiveness (8). However, these characteristics of muscle fibres may not always change (23,42) or may change in an undesirable direction, i.e. a decrease in fast, type IIX fibres, as a result of strength/power training (2,17). The large increase in explosive muscle performance in the present study may be related with a possible increase of the

relative amount of fast fibers (2), augmented by the greater protein synthesis during high-load eccentric exercise training (27). In addition, the large increase in maximal strength and rate of force development may also be due to increased neural adaptation from eccentric training, as previously shown by large increases in EMG activity after training (13,18,19). While EMG activity is lower during eccentric than concentric maximal contractions in single leg extension exercise (40,41), there is evidence to suggest that leg extensors are activated close to their maximum level when eccentric exercise is performed with multi-joint movements against high loads (29). In the present study, the increased ratio of maximal eccentric to concentric strength, from 93 before training to 113% after training, suggests that muscle activation may have increased. This may be due to enhanced neural activation as a result of training with high eccentric load, possibly caused by increased motor unit firing frequency and/or motor neuron recruitment, as previously suggested (1).

The finding that the ratio of maximal eccentric to concentric strength was low may be explained by mechanical factors affecting muscular force during the seated leg-press exercise, i.e. the force-length relationship of the muscles involved and the time to peak tension. According to unpublished data from our laboratory, the force applied on the force platform during the seated leg press is low at small knee angles, while it increases steeply from 120° to 155°. The eccentric muscle contractions in the present study started from a knee angle of $155 \pm 2^\circ$ and stopped at $126 \pm 1^\circ$, i.e. they were performed at the “steep” portion of the force vs. joint angle relationship, moving from a position of high to a position of low force generating capacity, according to the force-length relationship. In contrast, during the concentric contraction, the change in knee angle was from the smaller to the larger (i.e. towards knee extension), thus moving up on the ascending part of the force vs. joint angle relationship (i.e. towards a position of higher force generating

capacity). This, combined with the relatively slow time to attain peak force during the leg –press exercise, may explain the fact that eccentric peak force was lower before training.

The improved DJ performance and power, combined with the smaller changes in ankle, knee and hip angles (Fig. 2), indicated an increase in muscle stiffness during the DJ. Previous studies have shown that smaller changes in knee and hip angles during the landing phase of a drop jump are indicative of increased muscle stiffness that may be defined as the ratio of the change in joint moment and the change in joint angle. (36, 26). It has also been reported that performance efficiency during DJ is influenced by muscle-tendon stiffness during landing, which allows storage and recoil of elastic energy and the development of high muscle power (16). This apparent increase in muscle stiffness is related with the increase in eccentric strength and rate of force development (22), but may also be attributed to adaptations of cytoskeleton proteins such as titin that increase passive resistance of the muscle-tendon unit (38). In addition, Duclay et al. (12) reported a 16% increase in the stiffness of the gastrocnemius tendon following 18 sessions of eccentric training of the plantar flexors. Increased stiffness of the muscle-tendon unit due to eccentric training would, in turn, contribute to enhanced DJ performance. The relatively large increase in DJ performance in the present study with only a small volume of high intensity eccentric training may also be due to the movement specificity of the leg press exercise used. Strength training with closed-chain exercises has been shown to be more effective for improving vertical jump than open chain exercises (4). Similarly, strength measured using closed-chain exercises is highly correlated with DJ performance (6). Furthermore, the range of joint motion used during the eccentric leg press training in the present study was angle-specific for the drop jump movement, as indicated by the changes in leg joint angles (Fig. 2). Therefore, the

combination of eccentric high exercise intensity with the maximal loading throughout the range of motion (isokinetic contraction) and the angle specificity may explain the large gains in DJ performance.

One of the distinct characteristics of eccentric training is the phenomenon of muscle damage, mainly during the initial stages of training. Muscle damage is observed after vigorous, unaccustomed eccentric exercise and leads to muscle soreness due to structural disruptions of myofibrils and damage to the excitation-contraction coupling system (33). The inflammatory response that follows is considered to initiate a series of events leading to the release of growth factors that regulate satellite cell proliferation and muscle hypertrophy (32,37). In contrast with these positive effects of muscle damage, muscular strength and power are impaired for several days, depending on exercise intensity and volume, as well as the extent of muscle lengthening (32,33). However, when muscle damage occurs for the first time, a series of adaptations protect the muscle for 2-4 weeks, by reducing or preventing muscle damage in subsequent bouts; this is termed “repeated bout effect” (30). Interestingly, it has been shown that an eccentric contraction at slow velocity not only causes significantly less muscle damage, but it also reduces muscle damage in subsequent exercise bouts for at least 2 weeks (10,33). The design of the training program in the present study, including 3 familiarization sessions and a progressive increase in volume and intensity of eccentric exercise, minimized muscle damage and assisted the subjects to maintain high quality workouts throughout the training period.

In conclusion, the present study showed that large increases in both concentric and eccentric strength and explosiveness, as well as DJ performance can be attained with low volume (30-40 repetitions per session) high intensity eccentric exercise on this custom-made isokinetic

leg press machine. The maintenance of high resistance throughout the range of motion (isokinetic exercise mode), in combination with the visual feedback that assisted in maintenance of high effort, may partially explain the high efficiency of this training program.

PRACTICAL APPLICATIONS

The results of the present study show that our custom-made isokinetic leg press device may be considered as a highly efficient and safe means to increase concentric and eccentric strength and explosive performance of the legs. Due to the characteristics of eccentric loading, significant adaptations can occur with only 20-30 seconds of pure exercise time per training session, with the total of training time being less than 20-35 minutes. Thus, this type of low volume/high load eccentric training may be used by strength and conditioning coaches and trainers for fast gains in strength and power of the legs.

Furthermore, the seated position of the leg press exercise minimizes loading of the spine, thus allowing very high resistance to be applied safely on the leg muscles. This constitutes an important advantage of this training machine and it may be speculated that it could be used by coaches and clinicians to attain high eccentric loading of the leg muscles in a wide range of populations (e.g. injured, elderly, females). Due to the high muscle tension during the eccentric contractions, practitioners are cautioned to increase exercise volume and intensity progressively in order to minimize muscle damage. Visual feedback of force level was important for effective control of training load, and thus it is advised that visual feedback should be provided during high load eccentric strength training.

REFERENCES

1. Aagaard, P, Simonsen, EB, Andersen, JL, Magnusson, P, and Dyhre-Poulsen, P. Increased rate of force development and neural drive of human skeletal muscle following resistance training. *J Appl Physiol* 93: 1318-1326, 2002.
2. Andersen JL, Aagaard P. Effects of strength training on muscle fiber types and size; consequences for athletes training for high-intensity sport. *Scand J Med Sci Sports* 20 (Suppl 2): 32-38, 2010.
3. Arabatzi F, Kellis E. Olympic weightlifting training causes different knee muscle-coactivation adaptations compared with traditional weight training. *J Strength Cond Res* 26(8): 2192-2201, 2012.
4. Augustsson, J, Esko, A, Thomee, R, and Svantesson, U. Weight training of the thigh muscles using closed vs. open kinetic chain exercises: a comparison of performance enhancement. *J Orthop Sports Phys Ther* 27: 3-8, 1998.
5. Barr MJ, Nolte VW. The Importance of Maximal Leg Strength for Female Athletes When Performing Drop Jumps. *J Strength Cond Res*, DOI: 10.1519/JSC.0b013e31829999af. 2013.
6. Blackburn, JR, and Morrissey, MC. The relationship between open and closed kinetic chain strength of the lower limb and jumping performance. *J Orthop Sport Phys* 27: 430-435, 1998.
7. Bogdanis GC, Papaspyrou A, Souglis AG, Theos A, Sotiropoulos A, Maridaki M. Effects of two different half-squat training programs on fatigue during repeated cycling sprints in soccer players. *J Strength Cond Res* 25(7):1849-1856, 2011.
8. Campos GE, Luecke TJ, Wendeln HK, Toma K, Hagerman FC, Murray TF, Ragg KE, Ratamess NA, Kraemer WJ, Staron RS. Muscular adaptations in response to three different

- resistance-training regimens: specificity of repetition maximum training zones. *Eur J Appl Physiol* 88(1-2): 50-60, 2002.
9. Caruso JF, Hernandez DA, Saito K, Cho M, Nelson NM. Inclusion of eccentric actions on net caloric cost resulting from isoinertial resistance exercise. *J Strength Cond Res.* 17(3): 549-555, 2003.
 10. Chapman DW, Newton MJ, McGuigan MR, Nosaka K. Effect of slow-velocity lengthening contractions on muscle damage induced by fast-velocity lengthening contractions. *J Strength Cond Res* 25(1): 211-219, 2011.
 11. Cook CJ, Beaven CM, Kilduff LP. Three weeks of eccentric training combined with overspeed exercises enhances power and running speed performance gains in trained athletes. *J Strength Cond Res* 27(5): 1280-1286, 2013.
 12. Duclay, J, Martin, A, Duclay, A, Cometti, G, and Pousson, M. Behavior of fascicles and the myotendinous junction of human medial gastrocnemius following eccentric strength training. *Muscle Nerve* 39: 819-827, 2009.
 13. Enoka, RM. Eccentric contractions require unique activation strategies by the nervous system. *J Appl Physiol* 81: 2339-2346, 1996.
 14. Fleck, SJ, and Kraemer, WJ. Designing Resistance Training Programs. 3rd ed. Champaign, IL: *Human kinetics*, 2004.
 15. Harridge SD, Bottinelli R, Canepari M, Pellegrino MA, Reggiani C, Esbjornsson M, Saltin B. Whole-muscle and single fibre contractile properties and myosin heavy chain isoforms in humans. *Pflugers Archiv* 432: 913–920, 1996.
 16. Hoffren M1, Ishikawa M, Komi PV. Age-related neuromuscular function during drop jumps. *J Appl Physiol* 103: 1276-1283, 2007

17. Holm L, Reitelseder S, Pedersen TG, Doessing S, Petersen SG, Flyvbjerg A, Andersen JL, Aagaard P, Kjaer M. Changes in muscle size and MHC composition in response to resistance exercise with heavy and light loading intensity. *J Appl Physiol* 105(5):1454-1461, 2008.
18. Hortobagyi, T, Hill, JP, Houmard, JA, Fraser, DD, Lambert, NJ, and Israel, RG. Adaptive responses to muscle lengthening and shortening in humans. *J Appl Physiol* 80: 765-772, 1996.
19. Housh, TJ, Housh, DJ, Weir, JP, and Weir, LL. Effects of eccentric-only resistance training and detraining. *Int J Sports Med* 17: 145-148, 1996.
20. Isner-Horobeti ME, Dufour SP, Vautravers P, Geny B, Coudeyre E, Richard R. Eccentric exercise training: modalities, applications and perspectives. *Sports Med* 43(6): 483-512, 2013
21. Kraemer WJ, Ratamess NA. Fundamentals of resistance training: progression and exercise prescription. *Med Sci Sports Exerc* 36(4): 674-88, 2004
22. Kramer A, Ritzmann R, Gruber M, Gollhofer A. Four weeks of training in a sledge jump system improved the jump pattern to almost natural reactive jumps. *Eur J Appl Physiol* 112(1): 285-293, 2012
23. Kyrolainen H, Avela J, McBride JM, Koskinen S, Andersen JL, Sipilä S, Takala TE, Komi PV. Effects of power training on muscle structure and neuromuscular performance. *Scand J Med Sci Sports* 15(1): 58-64, 2005
24. Leszczak TJ, Olson JM, Stafford J, Brezzo RD. Early Adaptations to Eccentric and High-Velocity Training on Strength and Functional Performance in Community-Dwelling Older Adults. *J Strength Cond Res* 27(2): 442-448, 2013

25. Liu C, Chen CS, Ho WH, Füle RJ, Chung PH, Shiang TY. The effects of passive leg press training on jumping performance, speed, and muscle power. *J Strength Cond Res* 27(6): 1479-1486, 2013
26. Lyle MA, Valero-Cuevas FJ, Gregor RJ, Powers CM. Control of dynamic foot-ground interactions in male and female soccer athletes: Females exhibit reduced dexterity and higher limb stiffness during landing. *J Biomech.* 47(2): 512-517, 2014
27. Moore, DR, Phillips, SM, Babraj, JA, Smith, K, and Rennie, MJ. Myofibrillar and collagen protein synthesis in human skeletal muscle in young men after maximal shortening and lengthening contractions. *Am J Physiol Endocrinol Metab* 288: E1153-1159, 2005.
28. Nardone, A, Romano, C, and Schieppati, M. Selective recruitment of high-threshold human motor units during voluntary isotonic lengthening of active muscles. *J Physiol* 409: 451-471 1989.
29. Norrbrand, L, Fluckey, JD, Pozzo, M, and Tesch, PA. Resistance training using eccentric overload induces early adaptations in skeletal muscle size. *Eur J Appl Physiol* 102: 271-281, 2008.
30. Nosaka, K and Clarkson, PM. Muscle damage following repeated bouts of high force eccentric exercise. *Med Sci Sports Exerc* 27: 1263-1269, 1995.
31. Peterson, MD, Alvar, BA, and Rhea, MR. The contribution of maximal force production to explosive movement among young collegiate athletes. *J Strength Cond Res* 20: 867-873, 2006.
32. Philippou A, Papageorgiou E, Bogdanis G, Halapas A, Sourla A, Maridaki M, Pissimissis N, Koutsilieris M. Expression of IGF-1 isoforms after exercise-induced muscle damage in humans: characterization of the MGF E peptide actions in vitro. *In Vivo* 23(4): 567-575, 2009.

33. Proske, U, and Morgan, DL. Muscle damage from eccentric exercise: mechanism, mechanical signs, adaptation and clinical applications. *J Physiol* 537: 333-345, 2001.
34. Roig M, O'Brien K, Kirk G, Murray R, McKinnon P, Shadgan B, Reid WD. The effects of eccentric versus concentric resistance training on muscle strength and mass in healthy adults: a systematic review with meta-analysis. *Br J Sports Med* 43(8): 556-568, 2009
35. Saez-Saez De Villareal E., Kellis E., Kraemer W. J., and Izquierdo M. Determining variables of plyometric training for improving vertical jump height performance: a meta-analysis. *J Strength Cond Res* 23: 495-606, 2009.
36. Schmitz RJ, Kulas AS, Perrin DH, Riemann BL, Shultz SJ. (2007). Sex differences in lower extremity biomechanics during single leg landings. *Clin Biomech.* 22(6): 681-688, 2007.
37. Schoenfeld BJ. The mechanisms of muscle hypertrophy and their application to resistance training. *J Strength Cond Res* 24(10): 2857-2872, 2010
38. Toigo, M and Boutellier, U. New fundamental resistance exercise determinants of molecular and cellular muscle adaptations. *Eur J Appl Physiol* 97: 643–663, 2006.
39. Uh, BS, Beynnon, BD, Helie, BV, Alosa, DM, and Renstrom, PA. The benefit of a single-leg strength training program for the muscles around the untrained ankle. *Am J Sports Med* 28: 568-573, 2000.
40. Westing SH, Cresswell AG, Thorstensson A. Muscle activation during maximal voluntary eccentric and concentric knee extension. *Eur J Appl Physiol Occup Physiol* 62(2): 104-108, 1991

41. Westing SH, Seger JY, Thorstensson A. Effects of electrical stimulation on eccentric and concentric torque-velocity relationships during knee extension in man. *Acta Physiol Scand* 140(1): 17-22, 1990
42. Winchester JB, McBride JM, Maher MA, Mikat RP, Allen BK, Kline DE, McGuigan MR. (2008). Eight weeks of ballistic exercise improves power independently of changes in strength and muscle fiber type expression. *J Strength Cond Res* 22(6): 1728-1734, 2008
43. Winter, DA. Biomechanics and Motor control of Human Movement. Hoboken, NJ: John Wiley & Sons, Inc., 2009.

Table 1. Characteristics of the eccentric training program

Weeks	Sets	Repetitions	Intensity (% of maximal eccentric force)	Rest interval (min)	Total repetitions per week (sessions x sets x repetitions)
1	3	10	70	5	2 x 3 x 10 = 60
2	5	8	80	3	2 x 5 x 8 = 80
3	5	8	80	3	2 x 5 x 8 = 80
4	4	5	90	5	2 x 4 x 5 = 40
5	4	5	90	5	2 x 4 x 5 = 40
6	5	6	90	6	2 x 5 x 6 = 60
7	5	6	90	6	2 x 5 x 6 = 60
8	6	6	90	7	2 x 6 x 6 = 72
					Total = 492 reps

Table 2. Drop jump performance for the eccentric (ECC) and the control group (CG), measured before (PRE) and after (POST) training (mean $\pm$ SD).

Group	Variable	PRE	POST	Post vs. pre (%)	p value
ECC	DJ _{height} (cm)	34.2 $\pm$ 7.9	38.6 $\pm$ 8.1*	+13.6 $\pm$ 3.2%	0.001
	DJ _{ts} (msec)	318 $\pm$ 78	261 $\pm$ 70*	-17.6 $\pm$ 2.6%	0.001
	DJ _{pmax} (W)	4542 $\pm$ 520	5712 $\pm$ 645*	+25.8 $\pm$ 1.2%	0.001
CG	DJ _{height} (cm)	34.4 $\pm$ 8.2	34.1 $\pm$ 7.9	-0.7 $\pm$ 0.4%	0.99
	DJ _{ts} (msec)	324 $\pm$ 74	337 $\pm$ 89	+3.1 $\pm$ 1.8%	0.99
	DJ _{pmax} (W)	4692 $\pm$ 474	4532 $\pm$ 632	-3.7 $\pm$ 1.3%	0.94

DJ_{height}: drop jump height; DJ_{ts}: drop jump support time; DJ_{pmax}: drop jump maximal power

Table 3. Maximal eccentric force ($F_{\max\text{ECC}}$) and force attained in the first 300ms ($F_{300\text{ECC}}$) for the eccentric (ECC) and the control group (CG), measured before (PRE) and after (POST) training (mean $\pm$ SD). The p values are from the ANOVA post-hoc test for the interaction effect.

Group	Variable	PRE	POST	p value
ECC	$F_{\max\text{ECC}}$ (N)	3804 $\pm$ 694	6203 $\pm$ 942	0.001
	$F_{300\text{ECC}}$ (N)	2509 $\pm$ 638	3657 $\pm$ 612	0.001
CG	$F_{\max\text{ECC}}$ (N)	4121 $\pm$ 848	4063 $\pm$ 1140	0.98
	$F_{300\text{ECC}}$ (N)	2017 $\pm$ 620	1596 $\pm$ 890	0.42

Table 4. Maximal concentric force ($F_{\max\text{CON}}$) and force attained in the first 300ms ($F_{300\text{CON}}$) for the eccentric (ECC) and the control group (CG), measured before (PRE) and after (POST) training (mean $\pm$ SD). The p values are from the ANOVA post-hoc test for the interaction effect.

Group	Variable	PRE	POST	p value
ECC	$F_{\max\text{CON}}$ (N)	4212 $\pm$ 529	5503 $\pm$ 739	0.001
	$F_{300\text{CON}}$ (N)	1758 $\pm$ 778	3113 $\pm$ 732	0.001
CG	$F_{\max\text{CON}}$ (N)	4396 $\pm$ 998	4531 $\pm$ 1481	0.96
	$F_{300\text{CON}}$ (N)	1467 $\pm$ 426	1602 $\pm$ 461	0.63

Figure Legends

Fig.1. The custom made isokinetic leg press device

Fig. 2. Changes in ankle, knee and hip angle before (PRE) and after training (POST).

*: $p < 0.001$ from PRE, #: $p < 0.001$ from the POST value of the control group (CG)

ACCEPTED



Fig. 1

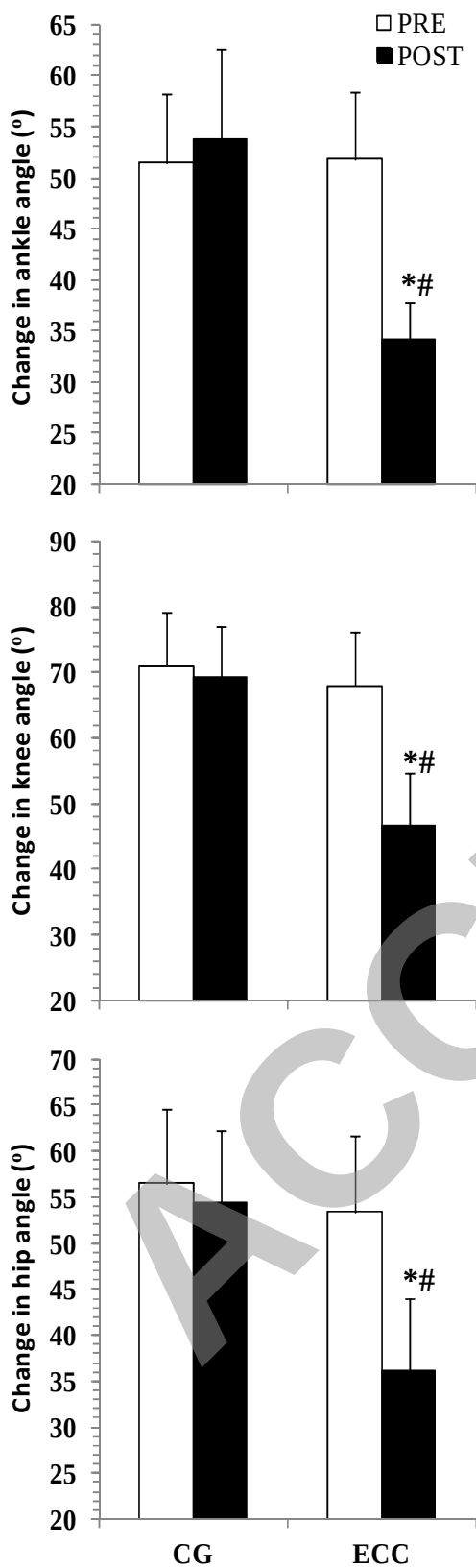


Fig. 2

